

ANONYMOUS AFFILIATIONS

- SUPPLEMENTARY MATERIAL -

Multi-Agent System
Description and Application

ANONYMOUS

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Abstract

METIS provides personalized academic support through a modular multi-agent system. Rather than assigning all tutoring functions to a single model, METIS distributes them among agents with distinct roles. The Manager Agent coordinates incoming requests, the Persona Agent maintains a representation of the learner, the Provider and Supporter Agents offer complementary forms of instructional assistance, and the Reminder Agent supports timely follow-up. These agents operate within a microservices architecture that separates the user interface, agent coordination, and data persistence. This paper presents the system architecture, coordination process, learner-modeling approach, and end-to-end processing of a student request.

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Chapter 1

Introduction

Large language models can produce fluent educational responses, but effective tutoring requires more than linguistic fluency. In a typical single-model system, one model must interpret the request, retrieve information about the learner, identify relevant course material, select an instructional strategy, and formulate the final response. Combining these functions within a single component makes pedagogical behavior difficult to control and limits the ability to determine why a response succeeds or fails.

METIS addresses this limitation by distributing these functions among cooperating agents. Coordination, learner modeling, explanation, problem-solving support, and follow-up remain connected, but each is handled as a distinct responsibility. The system is therefore designed not only to answer individual questions, but also to maintain continuity across tutoring interactions. Each response can account for the learner's academic background, recent activity, current difficulty, and preferred level of guidance.

This paper addresses four research questions:

1. How is METIS organized as a microservices-based learning system?
2. How does the Manager Agent coordinate the specialized agents?
3. How does the Persona Agent transform learner data into usable context?
4. How do the instructional agents balance explanation, scaffolding, and proactive support?

Chapter 2

Methodology

2.1 Overview of the METIS Architecture

METIS adopts a modular microservices architecture in which the main tutoring functions can be developed, tested, and evaluated separately. The architecture represents personalized tutoring as a set of identifiable responsibilities rather than a single opaque process. This separation supports analysis of how the interface, agent coordination, learner model, and persistent knowledge each contribute to the quality of academic support.

The system comprises three functional layers: the Interface and Interaction Layer, the Agentic Orchestration Layer, and the Knowledge Processing and Persistence Layer. Figure 2.1 presents the deployment structure and the flow of information among these layers.

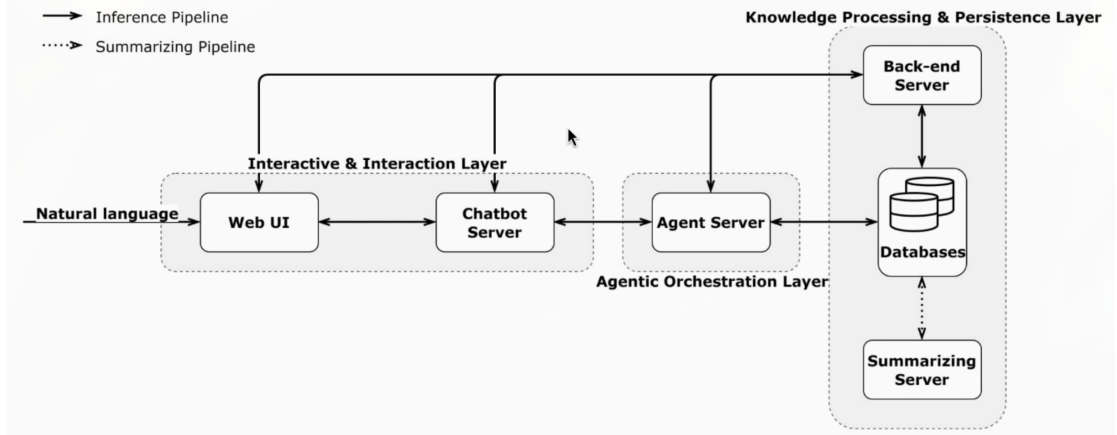


Figure 2.1: High-level architecture of the METIS microservices.

2.1.1 Interface and Interaction Layer

Students access METIS through the Interface and Interaction Layer. The Web UI provides a conversational workspace for submitting questions and receiving natural-language guidance. It can also provide access to profile settings, prior learning activity, reminders, and related course-support functions.

Requests from the Web UI are first routed to the Chatbot Server. The server validates the request format, performs basic preprocessing and sanitization, and associates the query with the appropriate session before forwarding it. As a result, the Agent Server receives a standardized request, allowing the tutoring agents to focus on instructional decisions rather than transport and formatting concerns.

2.1.2 Agentic Orchestration Layer

The Agentic Orchestration Layer is responsible for the principal instructional decisions within METIS. The Agent Server hosts the multi-agent framework and coordinates the Manager, Persona, Provider, Supporter, and Reminder Agents. It communicates with the Back-end Server, which manages authentication, session state, learner records, and data exchange between the user-facing application and the agent system.

When a standardized request reaches this layer, the system determines the appropriate form of assistance. Possible actions include explaining a concept, providing a sequence of hints, updating the learner model, scheduling a reminder, or combining several actions within one interaction.

2.1.3 Knowledge Processing and Persistence Layer

The Knowledge Processing and Persistence Layer grounds agent responses in course material and preserves information across sessions. Course outlines, institutional documents, and selected web resources are processed by the Summarizing Server, which extracts relevant content and converts it into structured summaries for subsequent retrieval.

The resulting summaries are stored in a MySQL database together with learner records and interaction histories. A shared database provides the agents with a consistent source of course context and allows learner information to persist across conversations. It also separates long-term storage from the agent processes that consume the data.

2.2 Knowledge Preparation and Synthesis

Before METIS can provide course-specific guidance, the source material must be converted into a representation that the agents can retrieve and interpret. This preparation process consists of five stages:

1. **Source acquisition:** Approved materials are collected, including course outlines, institutional pages, schedules, and learning objectives.
2. **Content extraction:** Digital documents and selected URLs are parsed into machine-readable text.
3. **Cleaning and normalization:** Navigation menus, repeated headers, formatting artifacts, and unrelated material are removed.
4. **Semantic summarization:** The Summarizing Server identifies learning objectives, prerequisite skills, course metadata, and important relationships between topics.
5. **Persistence:** The standardized records are saved to MySQL, where downstream services can retrieve a consistent version of the course context.

This pipeline provides a reliable curricular reference but does not determine the instructional strategy. Pedagogical decisions remain the responsibility of the instructional agents. The stored material instead provides the Provider and Supporter Agents with a consistent basis for tailoring explanations and guidance while maintaining alignment with the course.

2.3 Multi-Agent Framework and Orchestration Logic

The Agent Server hosts five agents, each assigned a distinct role in the tutoring process. Every request is coordinated by the Manager Agent, which invokes the other agents as required. Although multiple components may contribute to a response, the system presents their combined output as a coherent interaction. Figure 2.2 illustrates the delegation of tasks and the synthesis of agent outputs.

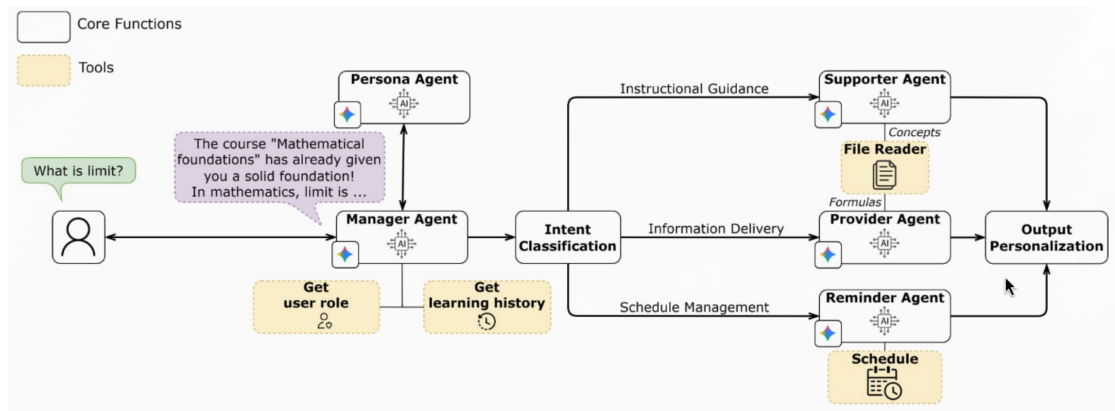


Figure 2.2: Interaction and task-delegation flow among METIS agents.

2.3.1 Manager Agent: Strategic Coordinator

The Manager Agent coordinates the response workflow from request analysis to final synthesis. After receiving a structured query, it performs four stages:

1. **Intent analysis:** The Manager identifies whether the student requires conceptual clarification, practical guidance, revision support, scheduling assistance, or another form of intervention.
2. **Context acquisition:** The Manager requests an updated learner representation from the Persona Agent before selecting an instructional response.
3. **Action planning and delegation:** The Manager selects the specialized agent, or combination of agents, most appropriate for the request.
4. **Output synthesis:** The Manager combines the returned results into a single response and calibrates its tone, level of detail, and proposed next step to the learner profile.

This process extends beyond keyword-based routing. The Manager evaluates the current request in relation to the student’s recent interaction history before selecting an instructional strategy. A conceptual question may be delegated to the Provider Agent, whereas difficulty progressing through an exercise may require the Supporter Agent. Requests concerning deadlines or revision planning may additionally invoke the Reminder Agent.

2.3.2 Persona Agent: Contextual and Behavioral Modeling

The Persona Agent integrates learner information into an operational student model that can be used by the other METIS agents. Rather than relying on a single undifferentiated profile, the agent draws on three sources: the Static Profile, Historical Context, and Realtime Query.

Static Profile

The Static Profile contains background information that is expected to remain relatively stable across sessions. This information may include enrolled subjects, prior academic performance, completed modules, faculty, major, and established relationships among previously studied topics. These attributes provide a baseline for estimating the learner’s prior knowledge and expected prerequisites.

Historical Context

Historical Context links the current request to previous interactions. It records recent discussions, previously encountered concepts, recurring errors, unresolved questions, and topics that remain in progress. This history enables continuity across sessions and prevents each request from being treated as an isolated interaction.

Realtime Query

The Realtime Query represents the student’s current request and immediate instructional need. In addition to its literal content, the query may indicate the

source of confusion, the learner’s intended outcome, the desired level of assistance, and relevant signs of frustration or uncertainty. Interpreting this information alongside the profile and interaction history helps METIS distinguish among requests for explanation, guided problem solving, revision support, and motivational assistance.

The Persona Agent combines these three sources and produces two forms of learner information:

- **User Context:** A short-term representation of the current interaction. It may include the active topic, recent learner feedback, emotional tone, patterns of success and failure, a summary of the latest exchanges, and a proposed next instructional step.
- **Dynamic Profile:** A longer-term representation inferred from repeated interactions. It may include learning preferences, rate of progress, proficiency level, mastered concepts, topics in progress, persistent difficulties, knowledge gaps, recurring misconceptions, and feedback preferences.

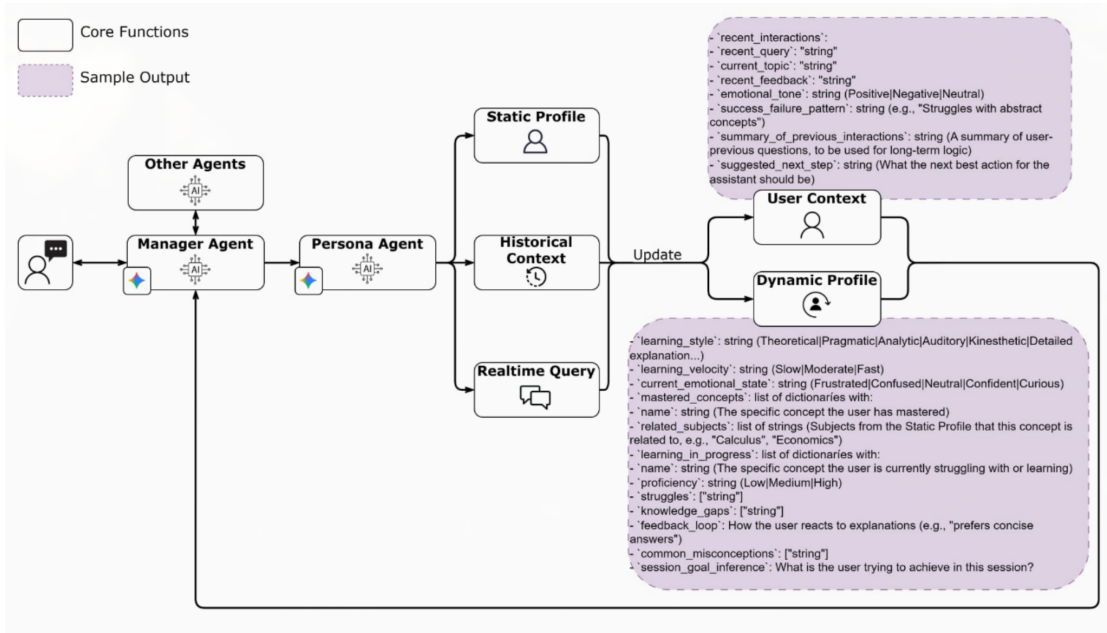


Figure 2.3: Persona Agent routing and learner-profile update mechanism.

Separating User Context from the Dynamic Profile prevents temporary conversational signals from being interpreted as persistent learner characteristics. This

separation also allows each downstream agent to receive only the information relevant to its role. The Manager can use short-term context to determine the next action, whereas the Provider and Supporter Agents can use the longer-term profile to calibrate explanation depth, hint progression, and the amount of instructional support.

2.3.3 Provider Agent: Personalized Knowledge Delivery

The Provider Agent handles requests that require direct knowledge delivery, including definitions, conceptual explanations, comparisons, examples, and clarification grounded in course material. Rather than producing a uniform response for all learners, the Provider adjusts its terminology, level of detail, and examples according to the student’s proficiency and current topic.

In particular, the Provider is expected to:

- retrieve the relevant course context from the persistent knowledge layer;
- choose a level of explanation suited to the learner profile;
- connect unfamiliar material to concepts the student already understands;
- address the underlying misconception instead of merely restating a definition; and
- offer a brief follow-up question or example when it would help check the student’s understanding.

2.3.4 Supporter Agent: Scaffolded Problem-Solving Guidance

The Supporter Agent assists students with applying knowledge while avoiding premature disclosure of a complete solution. It begins with the minimum level of guidance likely to support further progress. Depending on the learner’s difficulty, this guidance may take the form of a leading question, a relevant formula, a decomposition of the next step, or an analogous example.

The level of assistance can increase gradually over the course of the interaction. An initial response may identify the source of difficulty or provide a preliminary hint. If the learner remains unable to proceed, subsequent responses can reveal additional reasoning steps. This progression encourages independent problem solving

while reducing the likelihood of prolonged frustration.

2.3.5 Reminder Agent: Proactive Academic Intervention

The Reminder Agent enables METIS to provide proactive support rather than responding only to direct questions. It uses the Dynamic Profile, course milestones, and identified knowledge gaps to recommend appropriate future actions. These actions may include scheduling a review session, issuing an assignment reminder, prompting the student to revisit a weak topic, or notifying relevant participants about an academic event.

Each reminder includes a rationale for the proposed intervention. For example, METIS may recommend a review session on Friday afternoon because the student has repeatedly experienced difficulty with a prerequisite concept and has an assessment in the following week. Providing this rationale connects the reminder to the learner's academic needs rather than presenting it as a generic calendar notification.

2.4 End-to-End Agent Interaction Workflow

A typical interaction proceeds through the following sequence:

1. The student submits a query through the Web UI.
2. The Chatbot Server validates the input and converts it to a standardized format.
3. The Back-end Server associates the request with the appropriate session and learner identifiers.
4. Before selecting a response strategy, the Manager Agent requests an updated learner representation from the Persona Agent.
5. The Persona Agent combines the Static Profile, Historical Context, and Realtime Query, then returns the User Context and Dynamic Profile.
6. Based on this information, the Manager Agent develops an action plan and delegates the request to the appropriate agent or agents.
7. The selected agent retrieves course material when required and generates a role-specific output.

8. The Manager Agent synthesizes the outputs and adapts the final response to the learner profile.
9. The response and relevant updates to the learner state are stored for subsequent interactions.
10. The completed response is returned to the student through the Web UI.

2.5 Memory Management and Response Control

METIS organizes learner information across three time scales. The Static Profile captures stable academic background, the Dynamic Profile records patterns that develop over time, and the User Context represents the immediate conversational state. The system updates this learner representation before generating instructional content. This context-first procedure reduces the risk of basing a response on outdated or incomplete information.

Before a response is presented to the student, it can be evaluated against a defined set of operational constraints. Revision may be required if the response omits relevant profile information, exceeds the scope of the request, provides a complete solution when only a hint is appropriate, or conflicts with stored course material. The number of revision attempts is bounded to support correction without creating an uncontrolled execution loop.

The principal response controls are designed to:

- restrict each agent to its assigned role and authorized data sources;
- record the selected agent and action type to support traceability;
- prevent sensitive profile fields from appearing in the response;
- limit the number of retries after a response fails validation; and
- retain both the original student request and the approved response for subsequent analysis.

2.6 Implementation Availability

The implementation is divided into independent services, allowing individual components to be developed, tested, deployed, or replaced without redesigning the complete system. The source code is publicly available at https://github.com/htphung2000/personalized_assistant.git.

Chapter 3

Conclusion

METIS represents personalized tutoring as a coordinated set of responsibilities rather than the function of a single general-purpose model. The Manager Agent coordinates the response process, the Persona Agent maintains an evolving representation of the learner, the Provider and Supporter Agents offer direct explanation and scaffolded problem-solving assistance, and the Reminder Agent enables proactive follow-up. Together with persistent course and learner information, this architecture supports responses that remain aligned with the curriculum while adapting to the learner's developing needs across multiple interactions.